3.5 Scanners

3.5.1 ADC

Analogue-to-Digital Converter refers to the device employed to convert analogue signals into digital form. In a scanner the voltages (analogue) from the CCD are passed to the ADC to be converted into binary (digital format).

3.5.2 ADF

Automatic document Feeder is an attachment seen in some scanners that can load the pages that need to be scanned, automatically into the scanner. This allows scanning without any manual intervention.

3.5.3 Bit Depth

The amount of bits dedicated to capturing information from the scan subject. For greater detail, more bits need to be captured. A 24 bit depth allocates 8 bits or 1 byte per colour in RGB.

3.5.4 CCD

Charge Coupled Device refers to the optical sensor used to detect the intensity in the reflected light. These devices are capable of converting the light into electric energy with corresponding intensity. Higher intensity of light generates higher voltage electric impulse. For colour scanning the CCDs are coated with either of the colours Red, Green, Blue to register the colour under each filter.

3.5.5 Document Scanners

These are special type of Scanners that are designed for scanning large number of documents. Here rather than have a moving scan head, the scan head is fixed and the paper is fed underneath it.

A document scanner

3.5.6 Flatbed Scanners

These scanners have a sheet of glass over which the object to be scanned is placed. The scan head consisting